

Liberty Severs

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Brighton, United Kingdom.

PROFILE

I am a philosopher and cognitive scientist working on the topic of agency. I recently completed a resident fellowship at the Konrad Lorenz Institute for Evolution and Cognition Research in Vienna, Austria. My PhD thesis, based at the Faculty of Sciences at the University of Lisbon (March, 2023 – present) and the Institute for Philosophy II, Ruhr-Universität Bochum (March, 2024 – present), takes a theoretical and experimental approach to the comparative problem of agency. Here, I have been particularly interested in the comparative scope of reinforcement learning as a framework for operationalising agency and its functional variation across different species, aiming to provide an account of how constraints and divergent evolutionary histories shape the mechanisms and control processes that enable agential behaviour. Alongside my research, I am passionate about supporting interdisciplinary research communities, science communication, and science reform. I am a member of the PPI group for Interoception and Mental Health (University of Surrey), and serve as programme chair for the International Society for the Philosophy of the Sciences of the Mind (ISPSM). I am also co-founder of the Non-Neural Cognition Lab (nonneuralcognition.com), an interdisciplinary group studying agency and cognition beyond neural systems, which aims to improve the quality of research practices and scientific rigour of the field.

EXPERIENCE

- **Writing-up Fellow** October 2025 – Present
Konrad Lorenz Institute for Evolution and Cognition Research Vienna, Austria
 - Writing-up articles for a project titled 'Reinforcement learning and the Operational Foundations of Agency' [🌐]
 - Preparing thesis for submission.
- **Co-Founder & Research Lead, Non-Neural Cognition Lab** [🌐] December, 2024 – Present
Non-Neural Cognition Lab Bochum, Germany/Remote
 - Co-founded and coordinate an interdisciplinary research group focused on investigating agency and cognition in non-neural systems, especially Physarum polycephalum.
 - Develop and supervise experimental and theoretical research.
 - Lead collaborative projects and science communication initiatives.
- **PhD Student/Research Fellow** January 2023 – Present
University of Lisbon/Ruhr University Bochum Lisbon, Portugal/Bochum, Germany
 - PhD candidate (History and Philosophy of Science), interdisciplinary research on reinforcement learning and the organisation of agency across different species and systems.
 - Visiting Fellow, Institute for Philosophy II, Ruhr University Bochum.
- **Affect Researcher/Research Assistant** June 2022 – January 2023
Emteq Labs Brighton, UK
 - Medical technology-based research on emotion and cognition.
 - Involved in experimental research, robotics, VR/AR, and affective computing.
- **Research and Editorial Assistant** November 2021 – April 2022
Freelance RA/Editor Remote
 - Research and editorial support for Dr. Erik Hoel (Tufts) on a book about consciousness.

EDUCATION

- **University of Lisbon** March 2023 – April 2026 (expected)
PhD in History and Philosophy of Science Lisbon, Portugal
 - Dissertation: Reinforcement learning and the organisation of agency: a constraint-theoretic account
 - Supervisors: Dr. Davide Vecchi (UNED), Dr. Ines Hipolito (Macquarie University), Prof. Pedro Freitas (University of Lisbon)
- **University of Sussex** 2019 – 2021
MSc in Cognitive Neuroscience Brighton, UK
 - Dissertation: Intentional and reactive inhibitory control in Tourette Syndrome: a neuroimaging meta-analysis
 - Supervisor: Dr. Charlotte Rae (Sussex Centre for Consciousness Science)

- **University of Birmingham**
MA in Philosophy of Mind and Cognitive Science
◦ Dissertation: An evaluation of the ‘real problem’ approach to the science of consciousness
◦ Supervisor: Dr. Henry Taylor (University of Birmingham)
- **Heythrop College, University of London**
BA (Hons) in Philosophy
◦ Dissertation: Can Searle’s biological naturalism successfully account for consciousness?
◦ Supervisor: Dr. Michael Lacewing (UCL)

2017 – 2018
Birmingham, UK

2014 – 2017
London, UK

ADDITIONAL TRAINING

- Neuroanatomy, Brighton and Sussex Medical School/Sussex Neuroscience (2019)
- Computational Neuroscience, Neuromatch Academy (2021)
- Good Clinical Practice, NIHR (2022)

AWARDS AND GRANTS

- **Writing-up Fellow** 2025 - Present
Konrad Lorenz Institute for Evolution and Cognition
- **Visiting Fellow** 2024 - Present
Supervised by Tobias Schlicht and funded by the Volkswagen Foundation and RUB Research School, Institute for Philosophy II
- **International Research Exchange (Scholarship)** 2024 (6 months)
Ruhr University Bochum Research School
- **Doctoral Fellowship** 2023 (1 year)
Centre for the Philosophy of Science (CFCUL), University of Lisbon
- **Travel Grant** 2023
Italian Association for Cognitive Science (AISC), University of Genoa

ORGANISATION AND OUTREACH

- **Co-Founder, Non-Neural Cognition Lab** 2024–Present
<https://nonneuralcognition.com/>
- **Vice president, International Society for the Philosophy of the Sciences of the Mind** 2025–Present
Online
- **Programme chair, International Society for the Philosophy of the Sciences of the Mind** 2023–2024
Online
- **Organiser, 3rd International Conference for the Ethics of Artificial Intelligence** 2024
University of the Azores
- **Organiser, CELab PhD Seminar Series** 2023
CFCUL, University of Lisbon
- **Event Coordinator, Tolerance and Dogmatism in Science** 2023
CFCUL, University of Lisbon

TALKS AND PRESENTATIONS

- ***Physarum polycephalum* as a model organism for heterarchical control** Upcoming
Philosophy of Science Association (PSA), San Diego
- **Foraging decisions under resource uncertainty in *Physarum polycephalum*** April, 2026
Konrad Lorenz Institute for Evolution and Cognition Research (KL), Brown bag
- **Reinforcement and the organisation of agency** March, 2026
Konrad Lorenz Institute for Evolution and Cognition Research (KL), Brown bag
- **Beyond control: Reinforcement learning and the operational foundations of agency** November 2025
Konrad Lorenz Institute for Evolution and Cognition Research (KL), Fall Symposium
- **Organisation before representation: how control structures support adaptive strategies for learning and decision-making** October 2025
International Society for the Philosophy of the Sciences of the Mind (ISPSM) Annual Meeting

- **From local feedback to global goals: learning and decision-making in the acellular slime mould** June 2025
Reinforcement Learning and Decision Making (RLDM) Conference, Dublin, Ireland
- **Extended agency across ambient smart and design-centred environments** October 2024
International Society for the Philosophy of the Sciences of the Mind (ISPSM) Annual Meeting
- **Epistemic blindspots in the science of agency: the challenge of strange agents** September 2024
International Conference on the Ethics of Artificial Intelligence, University of the Azores
- **When the interoceptive and conceptual clash: Tourette syndrome and self-modelling under active inference** March 2024
European Society for Cognitive and Affective Neuroscience, Ghent University, Belgium
- **An active inference account of self-modelling in Tourette syndrome** January 2024
Theoretical Biology seminar, University College London
- **Strange agents and their implications for understanding biological and artificial systems** December 2023
Italian Association of Cognitive Sciences (AISC)
- **Embodiment and joint agency during human-robot interactions** August 2023
European Society for Philosophy and Psychology (ESPP)
- **Embodiment and joint agency during human-robot interactions** July 2023
Lisbon International Conference for the Philosophy of Science
- **The value of virtual and extended realities for situating agency** May 2023
XR Symposium and Summer School, University of Copenhagen
- **Multisensory modulation of the sense of self and presence in depersonalisation** March 2023
Champalimaud Foundation, Lisbon

PUBLICATIONS, PREPRINTS & THESIS MANUSCRIPTS

- 1 **Severs, L. & Wang, Q. (2025).** 'Continual decision-making dynamics across biological organisms' *Biological Reviews*.
- 2 **Becattini, V., & Severs, L. (2025).** Are ambient smart environments sufficient for developing good habits? *Synthese*, 206, 5.
- 3 **Parvizi-Wayne, D. & Severs, L. (2024).** When the interoceptive and the conceptual clash: The case of oppositional self-modelling in Tourette Syndrome. *Cognitive, Affective, & Behavioral Neuroscience*, 24, 660–680.
- 4 **Broulidakis, M. J., Kiprijanovska, I., Severs, L., Stankoski, S., Gjoreski, M., Mavridou, I., ... & Nduka, C. (2023).** Optomyography-based sensing of facial expression derived arousal and valence in adults with depression. *Frontiers in Psychiatry*, 14, 1232433.
- 5 **Severs, L. (2026).** Reinforcement learning and the organisation of agency: A constraint-theoretic account [Unpublished doctoral thesis].
- 6 **Wang, Q., Gaspers, N., Severs, L., Schlicht, T. (submitted).** Paired blue-light and nutrient exposure influences phototactic behaviour in *Physarum polycephalum*.
- 7 **Severs, L. (in preparation).** Agency and the interdependence of learning and decision-making strategies.
- 8 **Severs, L. (in preparation).** Control architectures and functions of agency.
- 9 **Severs, L., & Gaspers, N. (in preparation).** From local feedback to global goals: Learning and decision-making in the acellular slime mould.
- 10 **Severs, L., Gaspers, N., & Wang, Q. (in preparation).** Foraging decisions in *Physarum Polycephalum* under resource uncertainty.
- 11 **Severs, L. (in preparation).** The role of stochasticity in value-driven decision-making in neural and non-neural organisms.
- 12 **Severs, L. (in preparation).** Beyond control: Reinforcement learning and the operational foundations of agency.
- 13 **Severs, L., & Srećković, S. (in preparation).** On the question of deliberation with and without a brain.

TEACHING

- **Emotion and Rationality** 2025
Ruhr-Universität Bochum (Postgraduate seminar)

PODCASTS AND SCIENCE COMMUNICATION

- **'Interoception and self modelling in Tourette syndrome with Darius Parvizi-Wayne and Liberty Severs'** 2024
YouTube Podcast [YouTube](#)

PROFESSIONAL MEMBERSHIPS AND ROLES

- **International Society for the Philosophy of the Sciences of the Mind** *2024–present*
- **PPI Group for Interoception and Mental Health**, University of Surrey *2024–present*
- **European Society for Cognitive and Affective Neuroscience** *2024–present*
- **European Society for Philosophy and Psychology** *2023–present*

REFERENCE DETAILS

Email: tobias.schlicht@rub.de Prof. Tobias Schlicht, Institut für Philosophie II, Ruhr-Universität Bochum.

Email: cristina.villegas@kli.ac.at Dr. Cristina Villegas, Konrad Lorenz Institute for Evolution and Cognition Research.